

REGULATORY DOSSIER SUBMISSION

Dossier ID: DOS-C3941A859B

Country: Tanzania

Submission Date: 2025-05-09

Product: Lamivia (lamivudine) | ATC: J05AF10 | Strength: 150 mg

Applicant: National Therapeutics | Manufacturer: Gaborone Sterile Products

Policy Label: standard_review | Risk Score: 0.5

AMR Stewardship: AWaRe=not_applicable | Unmet Need=not_applicable | GLASS Trend=not_applicable
| Watch Similarity=not_applicable

SECTION CONTENT

[1] m1_application_admin - Application Form and Administrative Information

Labels: presence=present; length=length_ok; correctness=correct

This dossier constitutes a formal application for Marketing Authorization of Lamivia (lamivudine) submitted on 2025-05-09 to the national regulatory authority of Tanzania. The applicant, National Therapeutics, formally requests approval for the commercial distribution of this medicinal product. The proposed pharmaceutical form is tablet with a strength of 150 mg, classified under ATC code J05AF10. All required administrative declarations, legal attestations, letters of authorization, and regional application forms have been completed, signed by the responsible Qualified Person, and appended to Module 1. All referenced documents are available in the dossier annex and traceable by section identifier. The applicant confirms alignment with applicable technical guidance and regional submission standards. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Administrative cover letters, declarations, and fee records were cross-checked against the submission index. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The manufacturing site operates under a fully implemented

[1] m1_manufacturer_gmp - Manufacturer and GMP Evidence

Labels: presence=present; length=length_ok; correctness=correct

The primary commercial manufacturing site for the finished pharmaceutical product is Gaborone Sterile Products, located in Botswana. The facility was subject to a comprehensive GMP inspection by a recognized stringent regulatory authority. The latest inspection concluded on 2024-05-17, resulting in a formal compliance status of 'compliant'. The active GMP certificate number is GMP-750810, which remains valid until 2026-02-03. Site master files, a summary of recent inspection observations, and evidence of completed Corrective and Preventive Actions (CAPA) are included in the annex. Method validation records and quality controls were reviewed for internal consistency. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. All referenced documents are available in the dossier annex and traceable by section identifier. Recent regulatory inspections yielded no critical observations, and all major findings have been effectively closed. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Inspection history, CAPA closure records, and site responsibilities were reconciled across the manufacturing chain. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The applicant confirms alignment with applicable technical guidance and regional submission standards. Risk mitigation activities are docume

Site Inspection History Summary:

Inspection Date	Authority	Scope	Outcome
2024-05-17	National Auth	FPP/General	compliant
2022-05-12	SRA Joint	API/Sterile	compliant

[1] m1_product_information - Product Information and Naming

Labels: presence=present; length=length_ok; correctness=correct

The proposed proprietary name for this medicinal product is Lamivia, containing the active pharmaceutical ingredient lamivudine. The draft Summary of Product Characteristics (SmPC), Patient Information Leaflet (PIL), and primary/secondary packaging labels are provided for review. These documents detail the approved therapeutic indications, posology, contraindications, special warnings, and precautions for use. A comprehensive risk management plan to minimize medication errors is also submitted. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. All referenced documents are available in the dossier annex and traceable by section identifier. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Stability testing was conducted in climatic zones III a

[2] m2_quality_overall_summary - Quality Overall Summary

Labels: presence=present; length=length_ok; correctness=correct

This Quality Overall Summary (QOS) provides a critical evaluation of the chemistry, manufacturing, and controls (CMC) data for lamivudine. The control strategy justifies the proposed specifications for both the active substance and the finished product, emphasizing critical quality attributes (CQAs) and critical process parameters (CPPs). Impurity profiles, including degradation products and potential genotoxic impurities, have been thoroughly characterized. Batch analysis data from three commercial-scale validation batches demonstrate consistent manufacturing performance and compliance with all release criteria. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. The applicant confirms alignment with applicable technical guidance and regional submission standards. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. All referenced documents are available in the dossier annex and traceable by section identifier. Method validation records and quality controls were reviewed for internal consistency. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Data were analyzed using a mixed-effects model for r

[2] m2_clinical_overview - Clinical Overview and Benefit-Risk Summary

Labels: presence=present; length=length_ok; correctness=correct

The Clinical Overview synthesizes the efficacy and safety data supporting the use of Lamivia for the treatment of community acquired pneumonia. The pivotal clinical development program comprised 4 well-controlled studies. Based on the integrated analysis, the reported clinical outcome category is 'endpoint_not_met'. The overall benefit-risk profile is considered highly favorable for the target patient population. Safety findings were consistent with the known pharmacological class effects, and appropriate risk minimization measures have been integrated into the proposed prescribing information. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The clinical overview links the therapeutic rationale to the pivotal evidence package and the proposed indication. No new safety signals were identified during the extended open-label follow-up period. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Method validation records and quality controls were reviewed for internal consistency. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Benefit-risk language was reviewed for consistency with the submitted efficacy and safety summaries. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The target population demographics are broad

[3] m3_api_quality - API Quality and Control Strategy

Labels: presence=present; length=length_ok; correctness=correct

The active pharmaceutical ingredient (API) section details the complete synthesis route, starting from well-defined regulatory starting materials. Robust in-process controls and intermediate specifications ensure the consistent quality of the final drug substance. The analytical procedures used for release and stability testing have been fully validated for accuracy, precision, linearity, and robustness. The impurity control strategy adequately addresses organic impurities, residual solvents, and elemental impurities in accordance with current ICH guidelines. Forced degradation studies confirm the stability-indicating nature of the assay methods. Critical material attributes and control strategy elements were reviewed for consistency across the quality dossier. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The API package links specification limits to validated analytical methods and impurity-control decisions. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Method validation records and quality controls were reviewed for internal consistency. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Residual solvents and elemental impurities are controlled well below ICH Q3C and Q3D permissible daily exposures. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Statistical significance was set at an alpha

[3] m3_fpp_manufacturing - FPP Manufacturing Process and Controls

Labels: presence=present; length=length_ok; correctness=correct

The finished pharmaceutical product (FPP) manufacturing process utilizes standard, scalable unit operations. A formal quality risk assessment (e.g., FMEA) was conducted to identify critical process parameters (CPPs) that impact critical quality attributes (CQAs). Process performance qualification (PPQ) reports for three consecutive commercial-scale batches verify that the process is maintained in a state of control. In-process controls, intermediate hold-time studies, and packaging validation data are presented to substantiate the robustness and reproducibility of the commercial manufacturing operations. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. The applicant confirms alignment with applicable technical guidance and regional submission standards. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Method validation records and quali

[3] m3_stability - Stability and Shelf-Life Justification

Labels: presence=present; length=length_ok; correctness=correct

A comprehensive stability program has been executed to justify the proposed shelf-life and storage conditions. Data from both accelerated (40°C/75% RH for 6 months) and long-term storage conditions (spanning up to 36 months) are presented for multiple primary stability batches. Statistical evaluation using linear regression and poolability tests confirms that degradation trends remain well within the acceptable specification limits. Photostability and freeze-thaw cycling studies demonstrate that the product does not require specific handling precautions. A post-approval stability protocol commitment is included. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Method validation records and quality controls were reviewed for internal consistency. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Data were analyzed us

[4] m4_nonclinical_summary - Nonclinical Study Summary

Labels: presence=present; length=length_ok; correctness=correct

The nonclinical testing program evaluated the primary pharmacodynamics, secondary pharmacodynamics, safety pharmacology, and comprehensive toxicology profile of the compound. Repeat-dose toxicity studies in two mammalian species established clear no-observed-adverse-effect levels (NOAELs). Genotoxicity testing (Ames and in vivo micronucleus assays) yielded negative results. Reproductive and developmental toxicity studies revealed no teratogenic signals at clinically relevant exposures. The calculated safety margins support the proposed clinical dosing regimen without major safety concerns. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10.

All referenced documents are available in the dossier annex and traceable by section identifier. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Adverse events were coded using the Medical Dicti

[5] m5_trial_listing - Tabular Listing of Clinical Studies

Labels: presence=present; length=length_ok; correctness=correct

The tabular listing of clinical studies provides a comprehensive inventory of all trials conducted in support of the proposed indication (community acquired pneumonia). This includes Phase I pharmacokinetic/pharmacodynamic studies, Phase II dose-finding studies, and the pivotal Phase III efficacy and safety trials. For each study, the table details the protocol identifier, study design, randomization ratio, number of enrolled subjects, study duration, and current completion status. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Method validation records and quality controls were reviewed for internal consistency. The applicant confirms alignment with applicable technical guidance and regional submission standards. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Statistical significance was set at an alpha le

Pivotal Clinical Trial Inventory:

Protocol ID	Phase	Enrollment	Status
PROT-001	III	287	completed
PROT-002	III	541	completed
PROT-003	III	291	completed
PROT-004	III	721	completed

[5] m5_pivotal_trial_reports - Pivotal Clinical Trial Reports

Labels: presence=present; length=length_ok; correctness=correct

Detailed clinical study reports (CSRs) for the pivotal efficacy trials are provided. These randomized, double-blind, actively-controlled trials evaluated the primary and secondary endpoints in accordance with the pre-specified statistical analysis plan (SAP). The trials resulted in a formal outcome category of 'endpoint_not_met', demonstrating robust treatment effects in the intent-to-treat (ITT) and per-protocol populations. Subgroup analyses across age, gender, and baseline disease severity strata were consistent with the primary findings. Serious adverse events (SAEs) and discontinuations were systematically analyzed and adjudicated. The trial reports connect efficacy conclusions to the prespecified analysis framework and observed safety findings. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The analytical procedure was validated in

accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Method validation records and quality controls were reviewed for internal consistency. The primary efficacy endpoint was met, demonstrating a statistically significant improvement over placebo ($p < 0.001$). The pivotal study narratives align endpoint definitions, analysis populations, and outcome interpretation. Clinical study reports document protocol adherence, endpoint handling, and safety interpretation in the target population. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. All referenced documents are available in the dossier annex and traceable by section identifier. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. The applicant confirms alignment with applicable technical guidance and regional submission standards. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Discontinuations due to treatment-emergent adverse events (TEAEs) were low and balanced across all study arms. Adverse events were coded using the Medical Diction

[5] m5_bioequivalence - Biopharmaceutics and Bioequivalence Evidence

Labels: presence=present; length=length_ok; correctness=correct

Biopharmaceutics data substantiate the formulation development bridging between clinical trial materials and the proposed commercial product. In vivo bioequivalence studies demonstrated that the 90% confidence intervals for the geometric mean ratios of C_{max} and AUC parameters fell entirely within the standard acceptance criteria of 80.00% to 125.00%. Additionally, multi-point comparative dissolution profiles across physiological pH ranges (pH 1.2, 4.5, and 6.8) confirmed similarity ($f_2 > 50$). The bioanalytical methods used for plasma quantification were fully validated. Method validation records and quality controls were reviewed for internal consistency. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. All referenced documents are available in the dossier annex and traceable by section identifier. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The risk management plan includes routine